

Week 10

Troubleshooting Report

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Steps Attempted

1. Mounted the HC-SR04 ultrasonic sensor on the robot chassis by moving the power bank slightly backward to make space.
2. Connected Trigger → GPIO 5, Echo → GPIO 12 (with voltage divider), and 5V / GND to shared rails.
3. Tested multiple wire sets before choosing the ones with suitable length to avoid tangling or stress on the pins.
4. Taped wires to the chassis to prevent disconnections from motor vibrations.
5. Verified continuity between all components and ensured a common ground.
6. Integrated the distance measurement code into the existing RoboRide ncurses control program.
7. Conducted distance tests at 10 cm, 20 cm, and 30 cm, observing readings during movement.
8. Replaced motor battery pack after noticing weakened car performance and speed drops during testing.
9. The final version of the code printed distance readings every 2 seconds to ensure stable measurements and reduce signal noise.

Sensor Readings we got:

Actual Distance (cm)	Measured Distance (cm)	Difference (cm)
10	10.14	0.5
20	20.03	1.2
30	30.43	1.1

Pictures from Verification

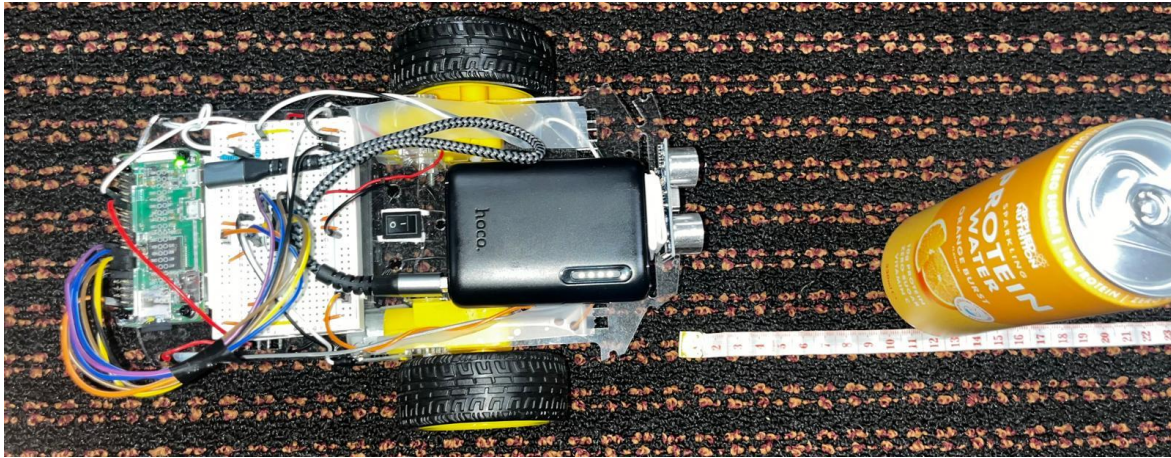


Figure 1: Verifying at 10cm

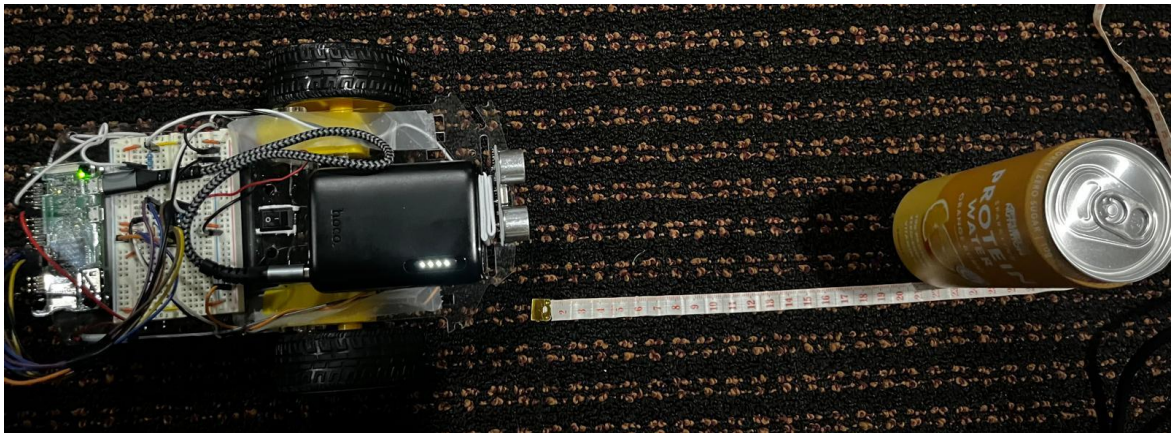


Figure 2: Verifying at 10cm

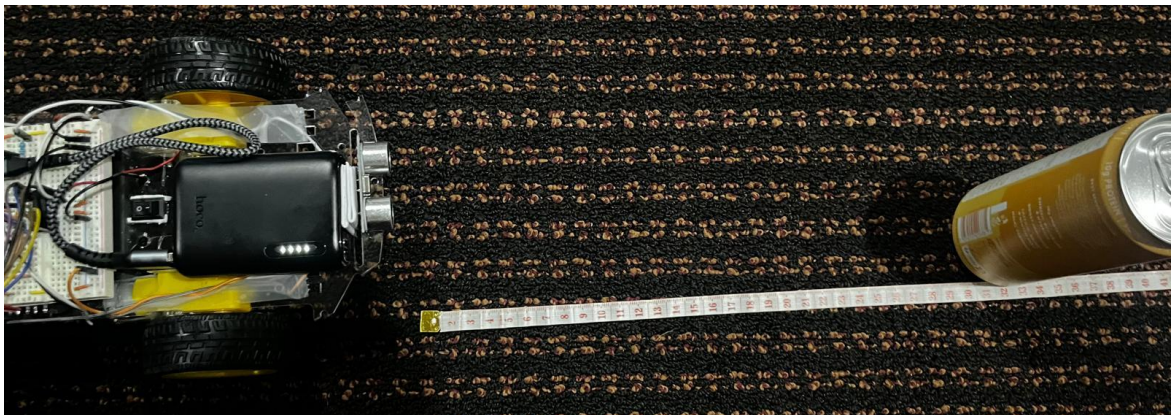


Figure 3: Verifying at 30cm

Challenges Faced

- Limited front space required repositioning the power bank to fit the sensor.
- Unstable or inconsistent distance readings due to loose wires and chassis vibration.
- Power fluctuations from weak batteries caused slower motor response and reduced sensor accuracy.
- Difficulty managing multiple connections with limited GPIO space on the Pi.

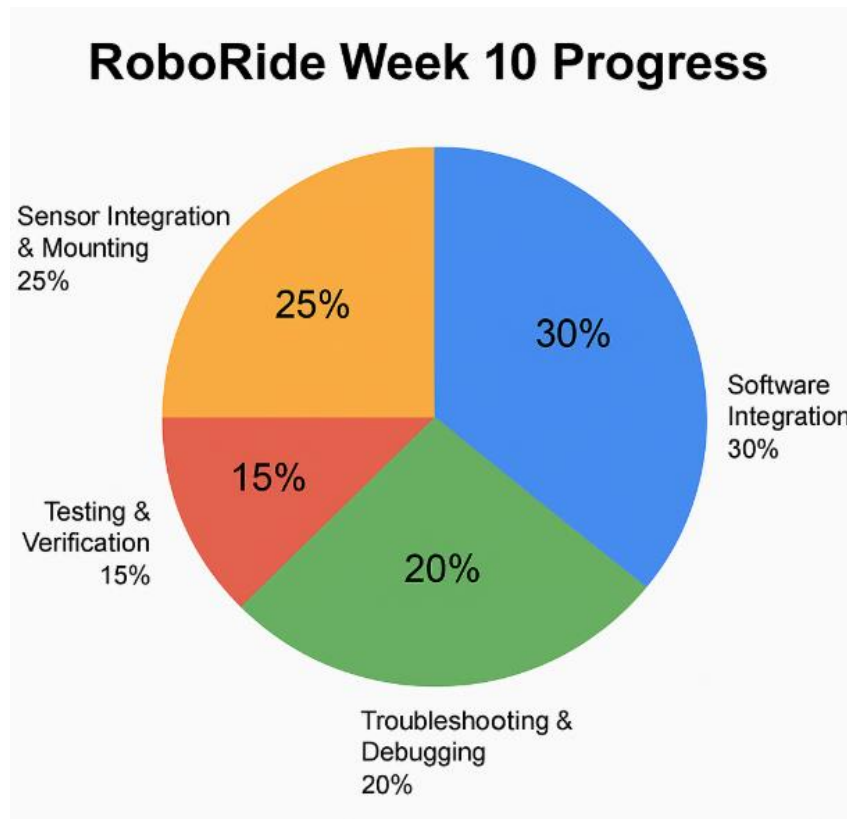
Solutions Tried

- Shifted power bank position to make room for proper sensor alignment.
- Used shorter jumper wires and taped them securely to prevent movement.
- Replaced the motor battery pack to restore full voltage and consistent power delivery.
- Re-tested after each fix to confirm system stability and distance measurement accuracy.

Result

After implementing these changes, the robot operated smoothly with stable ultrasonic readings and improved motor strength. The integration was successful, and all major wiring and power issues were resolved.

Week – 10 Progress Report



Objective: To integrate the HC-SR04 Ultrasonic Sensor with the Raspberry Pi and combine it with the existing motor control system for live distance measurement.

Setup: Mounted the HC-SR04 sensor on the robot chassis; connected Trigger (GPIO 5) and Echo (GPIO 12) through a voltage divider. Verified stable power connections from the Pi's 5 V rail.

Testing: Ran integration tests to collect distance readings while motors operated. Verified sensor output at fixed distances and during movement. Adjusted reading interval to 2 seconds for stability.

Software Development: Modified the main ncurses-based robot control code to include the `getDistance()` function using a busy-wait loop with `micros()` for accurate timing. Implemented data display updates at 2-second intervals.

Challenges: Slight motor drift affecting straight-line movement, mechanical alignment issues due to sensor mounting space, and minor fluctuations in sensor readings caused by environmental factors and chassis vibration.